Elsa Puspa Silfia

I am a Master's student in Mathematics in Science and Engineering at TUM with research experience in spatial transcriptomics, machine learning, and computational biology. During my thesis at Helmholtz Munich, I studied treatment-associated changes in cellular composition and gene expression in obesity experiments. I am particularly interested in single-cell and spatial transcriptomics data, perturbation modeling, and multimodal integration to develop predictive models of cellular response.

Education

Mar 2023 – Present **M.Sc. Mathematics in Science and Engineering**,
Technical University of Munich (TUM), Germany,
Courses: Machine Learning, AI in Medicine, Foundations of Data Analysis, Mathematical Theory of Foundation Models
Master's Thesis: Spatial Niche Detection and Treatment Effects Quantification in Xenium 5k Data, in collaboration with Helmholtz Munich, Institute of Computational Biology (Theis Lab)

Aug 2012 – Oct 2016 **B.Sc. in Mathematics**,
Bandung Institute of Technology, Indonesia
Thesis: Longevity Bond Valuation as a Hedging Instrument for Life Insurance and Pension Fund with Stochastic Shock of Mortality (GPA: 3.46/4.00)

Professional Experience

Oct 2025 – May 2026 **Master's Thesis Student**, Helmholtz Munich, Inst. for Computational Biology (Theis Lab) .
- Analyzed spatial transcriptomics data to identify cellular niches and quantify treatment effects in Xenium 5k data
- Applied representation learning and deep-learning-based niche modeling to characterize region-specific and niche-specific transcriptional responses
- Works in a High Performance Computing (HPC) environment using Linux and Git for collaboration.

Mar 2022 – May 2024 **Data Analyst**, Evermos.
- Designed and maintained SQL-based analytical datasets and automated reporting pipelines using SQL, dbt, and Airflow. Used Git/GitHub for version control, code collaboration, and reviewing changes within the data team.
- Developed dashboards visualizations in Metabase to support business monitoring and decision-making.
- Performed exploratory analysis and A/B testing using Python to evaluate business initiatives.
- Collaborated with business and engineering teams to improve data quality by validating datasets and investigating reporting inconsistencies.
- Led SQL and dashboard workshops for more than 50 colleagues, promoting self-service analytics.

Nov 2018 – Jan 2022 **Market Risk Management Specialist**, CIMB Niaga Bank.
- Developed and applied time series models to analyze market risk and support daily risk monitoring and decision-making
- Built automated workflows for recurring risk reports, improving process efficiency and reducing repetitive manual work
- Initiated and developed an anomaly detection approach for foreign exchange transactions using dynamic thresholds derived from time series forecasting
- Presented the anomaly detection initiative to the Director of Risk Management, where it was selected as a finalist in an internal innovation program

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Dec 2016 – Nov 2018 **Research Fellow**, Department of Macroprudential Policy, Central Bank of Indonesia.

- Developed quantitative models to assess systemic risk in the banking sector using large-scale financial and macroeconomic data
- Conducted stress testing and scenario-based risk analysis under different economic conditions
- Built automated reporting tools and dashboards linked directly to databases, reducing regulatory reporting time from days to near real-time
- Processed and analyzed large datasets to support reporting and insights for senior policymakers

Skills

Programming	Python, SQL, PyTorch, TensorFlow, Scanpy, scvi-tools
Data Eng.	Airflow, dbt, Snowflake, Data Pipelines, Data Modeling, Git/GitHub, Linux, HPC
Analytics & ML	Data Visualization (Metabase), Statistical Analysis, A/B Testing, Machine Learning, Deep Learning

Research & Academic Projects

AI in Medicine Project — Stroke MRI Anomaly Detection

- Developed diffusion-based generative models for pseudo-healthy brain MRI reconstruction and anomaly detection and evaluated lesion localization performance under different noise settings in medical imaging

Seminar Project — Transformers Learn In-Context by Gradient Descent: Analysis and Perspectives

- Studied recent research on transformer-based foundation models and in-context learning, focusing on gradient-descent interpretations of transformer behavior.

Scientific Computing Project — Dynamic Deformation of an Elastic Robot Fingertip

- Simulated static and dynamic elastic contact with friction using the Finite Element Method (FEM), employing ANSYS and FEniCS to analyze numerical solutions under different contact conditions.

Achievements

2026

- Young Researcher - Heidelberg Laureate Forum 2026

2021

- Shopee Code League – Top 8% (Product Detection Challenge)
- Shopee Code League – Top 12% (NLP Challenge)
- BRI Data Hackathon – Top 14% (Cash Ratio Optimization Challenge)

2016

- 2nd Prize – OJK National Paper Competition
- Honorable Mention – COMAP Mathematical Contest in Modeling (MCM)

2015

- 3rd Prize – Bank Indonesia Financial Stability Competition
- Meritorious Winner – COMAP Mathematical Contest in Modeling (MCM)

Languages

Indonesian (native), English (fluent), German (basic)